

D 120187

(Pages : 2)

Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION, MARCH 2025**

Mathematics

MTS 6B 14 (E01)—GRAPH THEORY

(2019 Admission onwards)

Time : Two Hours

Maximum : 60 Marks

**Part A***Answer any number of questions.**Each question carries 2 marks.**Ceiling is 20.*

1. Define complete graph. Draw  $K_6$ .
2. Draw all non isomorphic simple graphs of order 4.
3. How many edges are there for a  $k$ -regular graph of order  $n$  ?
4. Differentiate proper subgraphs and spanning subgraphs.
5. State Cayley's theorem on spanning trees of a complete graph.
6. Define vertex connectivity of a graph. Find the vertex connectivity of the cycle  $C_8$ .
7. State Whitney's theorem for 2-connected graphs.
8. Distinguish between bridge and cut vertex of a graph with example.
9. Find the Hamiltonian closure of the cycle  $C_4$ .
10. Give a characterization of Eulerian graphs. Show that  $K_5$  is Euler.
11. Define a maximal non-Hamiltonian graph.
12. If  $e$  is an edge of the complete graph  $K_5$ , then illustrate that  $K_5 - e$  is planar.

**Turn over**

**Part B**

*Answer any number of questions.*

*Each question carries 5 marks.*

*Ceiling is 30.*

13. Prove that, in any graph  $G$ , the number of vertices of odd degree is even.
14. Define union and intersection of two graphs. Explain with example.
15. Prove that any tree  $T$  with at least two vertices has more than one vertex of degree 1.
16. Prove that a connected graph  $G$  with  $n$  vertices has at least  $n - 1$  edges.
17. Let  $v$  be a vertex of the connected graph  $G$ . Then prove that  $v$  is a cut vertex of  $G$  if and only if there are two vertices  $u$  and  $w$  of  $G$ , both different from  $v$ , such that  $v$  is on every  $u - w$  path in  $G$ .
18. Prove that a simple graph  $G$  is Hamiltonian if and only if its closure  $c(G)$  is Hamiltonian
19. Prove that  $K_{3,3}$  is nonplanar.

**Part C**

*Answer any **one** question.*

*The question carries 10 marks.*

20. (i) Explain adjacency matrix and incidence matrix of a graph  $G$  with an example.  
(ii) Given any two vertices  $u$  and  $v$  of a graph  $G$ , prove that every  $u - v$  walk contains a  $u - v$  path.
21. Define spanning tree. Prove that a graph  $G$  is connected if and only if it has a spanning tree.

(1 × 10 = 10 marks)